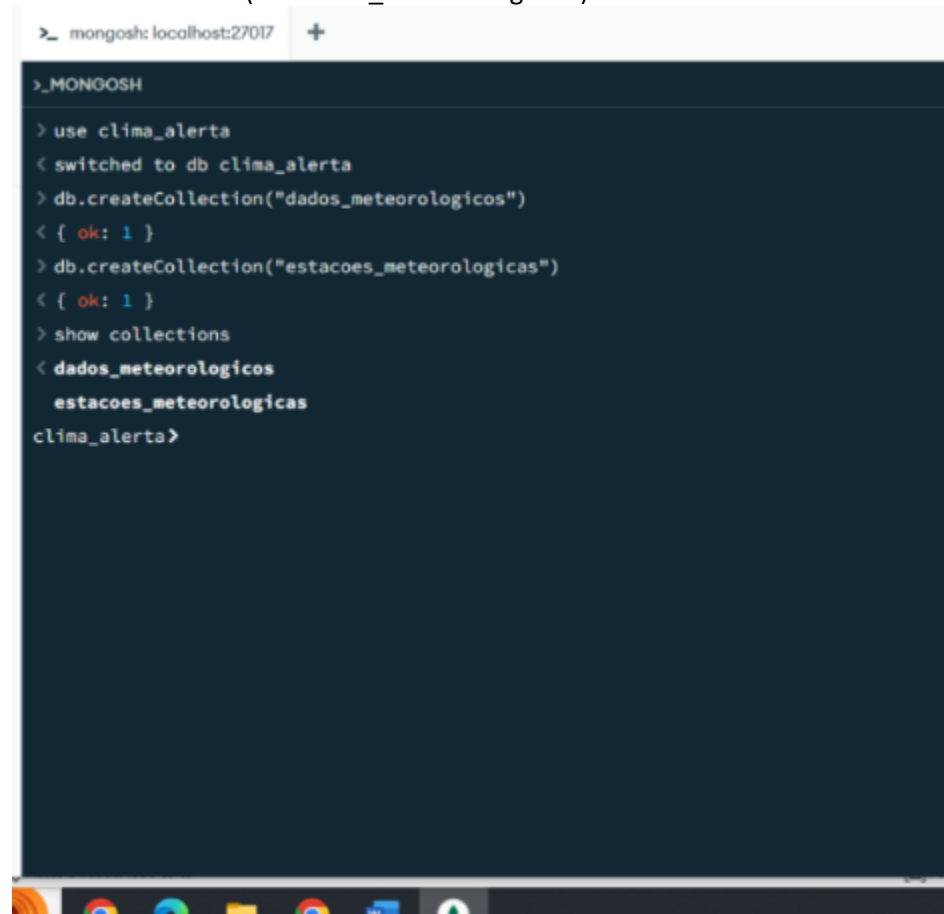


1)

```
use clima_alerta
```

```
db.createCollection("dados_meteorologicos")
```

```
db.createCollection("estacoes_meteorologicas")
```



The screenshot shows a terminal window titled "mongosh: localhost:27017". The prompt is ">\_MONGOSH". The user enters the following commands and receives the following responses:

```
> use clima_alerta
< switched to db clima_alerta
> db.createCollection("dados_meteorologicos")
< { ok: 1 }
> db.createCollection("estacoes_meteorologicas")
< { ok: 1 }
> show collections
< dados_meteorologicos
   estacoes_meteorologicas
clima_alerta>
```

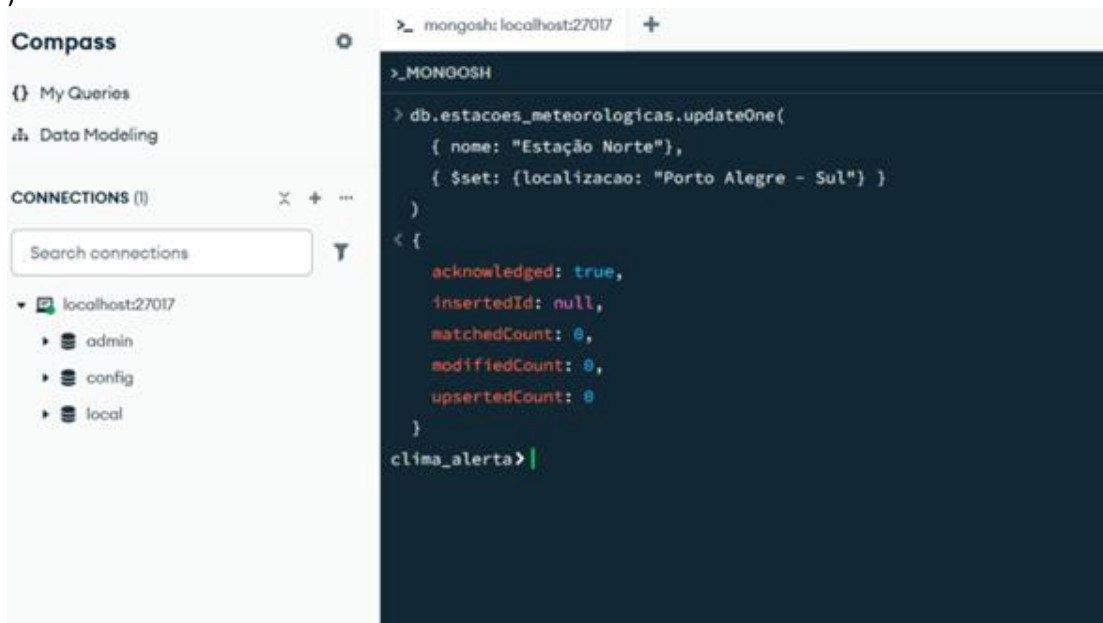
2)

```
db.estacoes_meteorologicas.updateOne(
```

```
{ nome: "Estação Norte"},
```

```
{ $set: {localizacao: "Porto Alegre - Sul"}} }
```

)



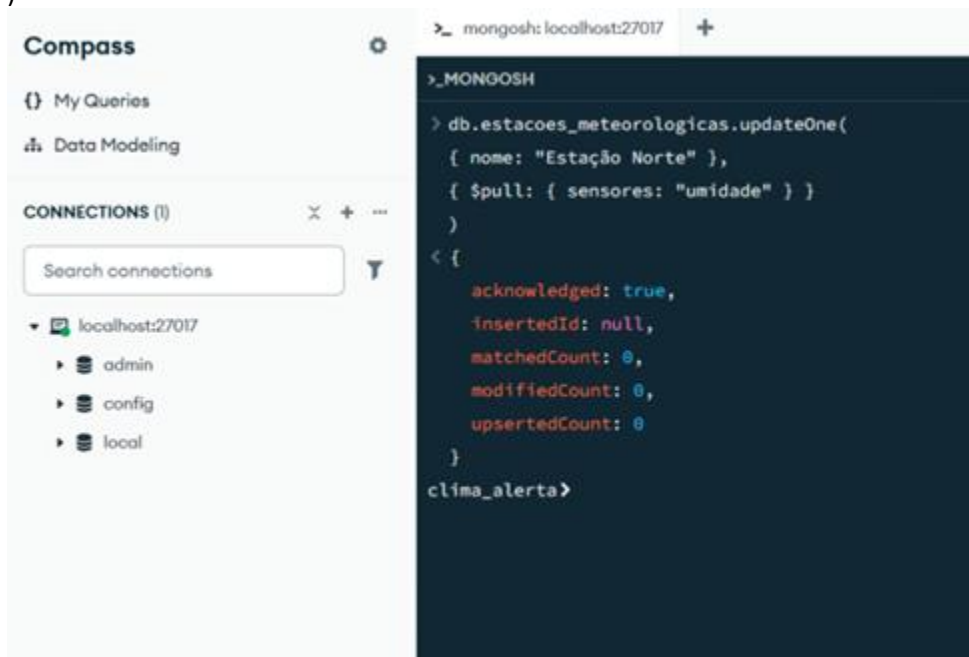
3)

```
db.estacoes_meteorologicas.updateOne(
```

```
{ nome: "Estação Norte" },
```

```
{ $pull: { sensores: "umidade" } }
```

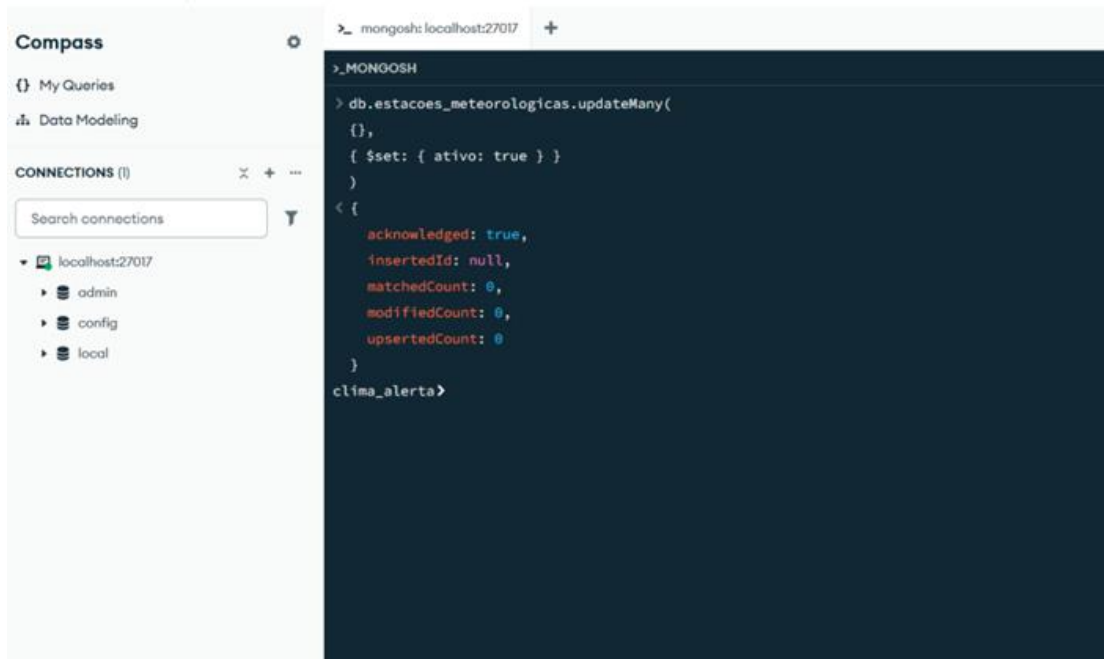
)



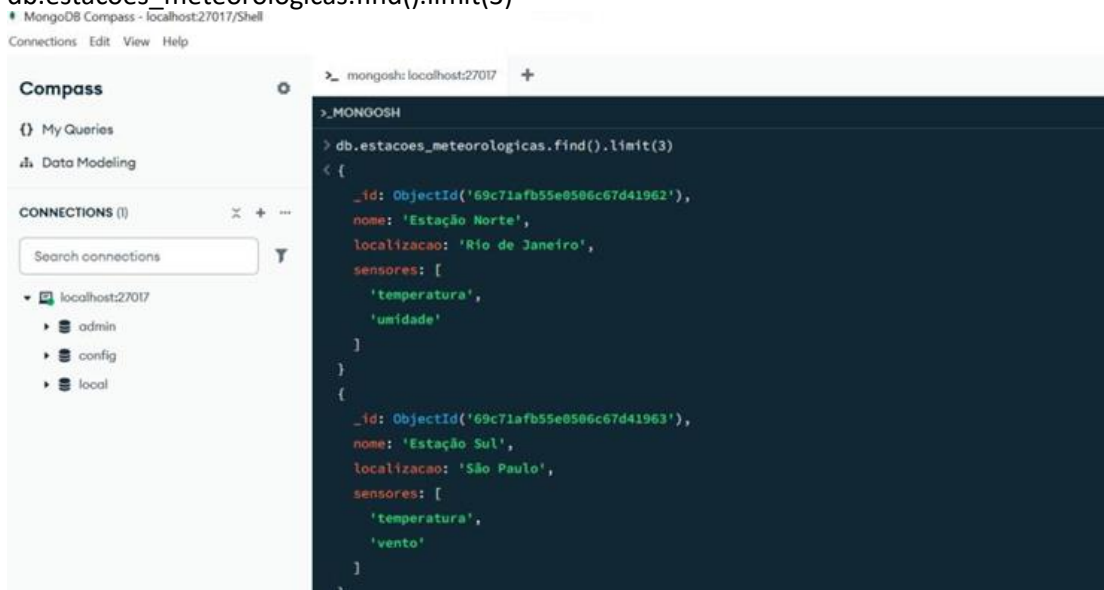
4)

```
db.estacoes_meteorologicas.updateMany(
```

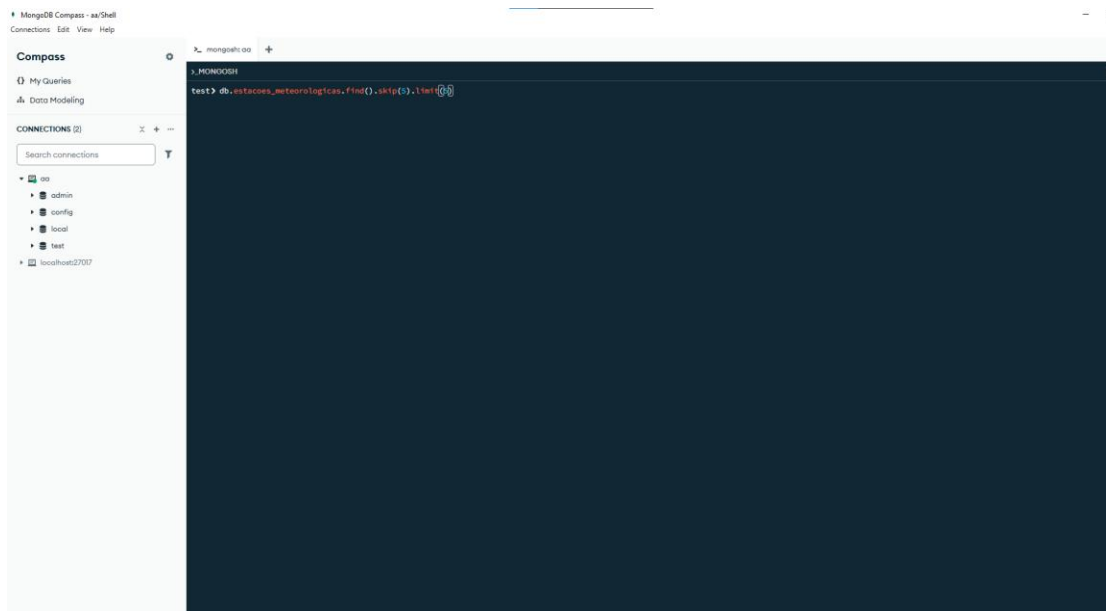
```
{},  
{ $set: { ativo: true } }  
)
```



5)  
`db.estacoes_meteorologicas.find().limit(3)`

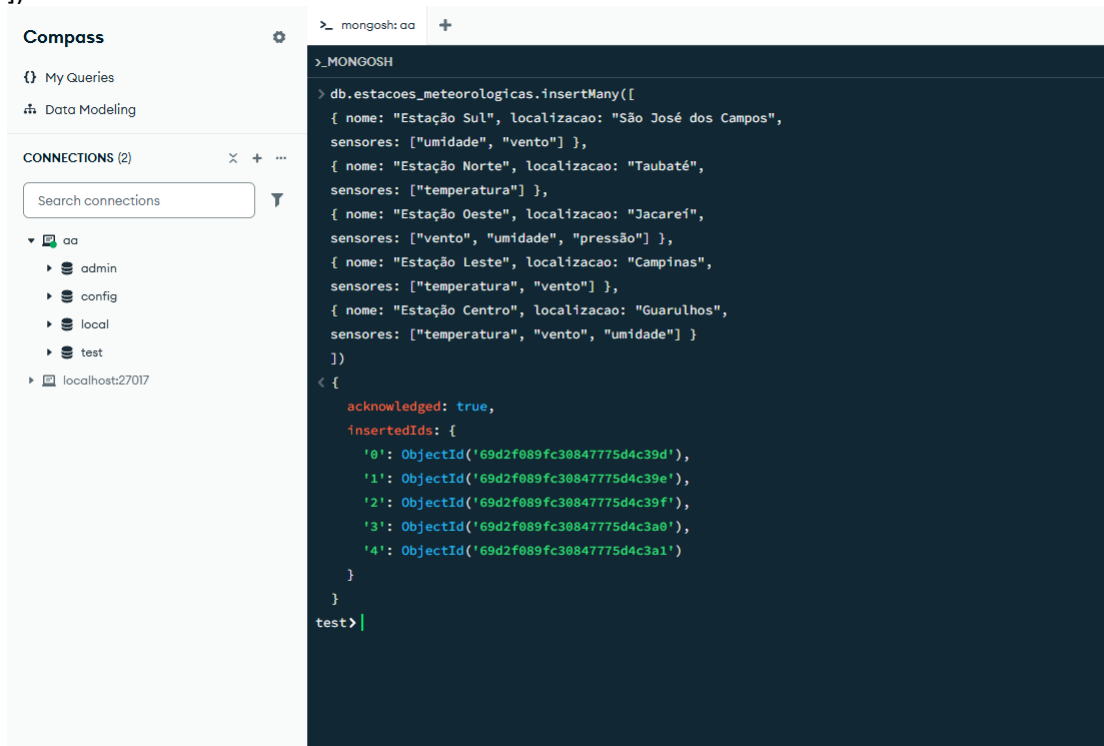


6)  
`db.estacoes_meteorologicas.find().skip(5).limit(5)`



```
7)
db.estacoes_meteorologicas.insertMany([
{ nome: "Estação Sul", localizacao: "São José dos Campos",
sensores: ["umidade", "vento"] },
{ nome: "Estação Norte", localizacao: "Taubaté",
sensores: ["temperatura"] },
{ nome: "Estação Oeste", localizacao: "Jacareí",
sensores: ["vento", "umidade", "pressão"] },
{ nome: "Estação Leste", localizacao: "Campinas",
sensores: ["temperatura", "vento"] },
{ nome: "Estação Centro", localizacao: "Guarulhos",
sensores: ["temperatura", "vento", "umidade"] }
```

})



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (2)' panel lists connections: 'aa' (selected), 'admin', 'config', 'local', 'test', and 'localhost:27017'. The main terminal window, titled 'mongosh: aa', shows the following command and output:

```
> db.estacoes_meteorologicas.insertMany([
  { nome: "Estação Sul", localizacao: "São José dos Campos",
    sensores: ["umidade", "vento"] },
  { nome: "Estação Norte", localizacao: "Taubaté",
    sensores: ["temperatura"] },
  { nome: "Estação Oeste", localizacao: "Jacareí",
    sensores: ["vento", "umidade", "pressão"] },
  { nome: "Estação Leste", localizacao: "Campinas",
    sensores: ["temperatura", "vento"] },
  { nome: "Estação Centro", localizacao: "Guarulhos",
    sensores: ["temperatura", "vento", "umidade"] }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69d2f089fc30847775d4c39d'),
    '1': ObjectId('69d2f089fc30847775d4c39e'),
    '2': ObjectId('69d2f089fc30847775d4c39f'),
    '3': ObjectId('69d2f089fc30847775d4c3a0'),
    '4': ObjectId('69d2f089fc30847775d4c3a1')
  }
}
test> |
```

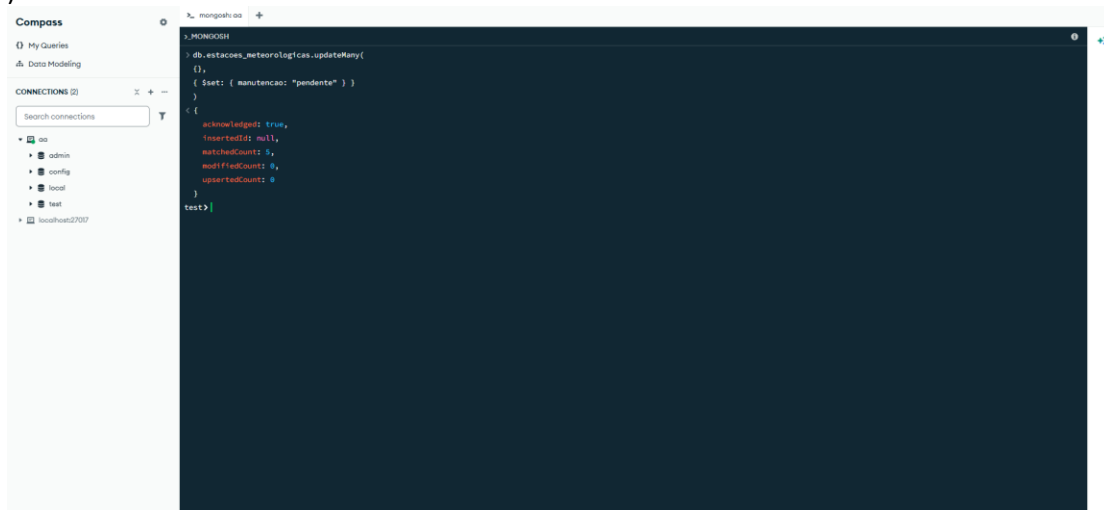
8)

db.estacoes\_meteorologicas.updateMany(

{},

{ \$set: { manutencao: "pendente" } }

)



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (2)' panel lists connections: 'aa' (selected), 'admin', 'config', 'local', 'test', and 'localhost:27017'. The main terminal window, titled 'mongosh: aa', shows the following command and output:

```
> db.estacoes_meteorologicas.updateMany(
  {},
  { $set: { manutencao: "pendente" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
test> |
```

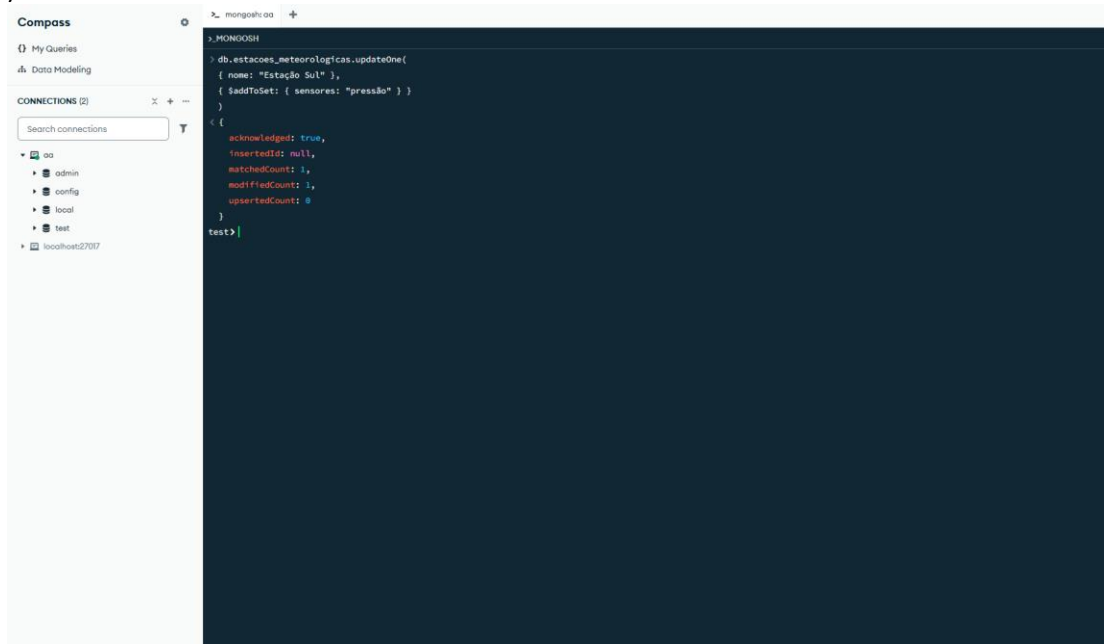
9)

db.estacoes\_meteorologicas.updateOne(

{ nome: "Estação Sul" },

```
{ $addToSet: { sensores: "pressão" } }
```

```
)
```



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel shows a connection to 'localhost:27017'. The main editor displays a JavaScript snippet for an update operation. The output shows the result of the operation, indicating that one document was updated successfully.

```
> db.estacoes_meteorologicas.updateOne(
  { nome: "Estação Sul" },
  { $addToSet: { sensores: "pressão" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

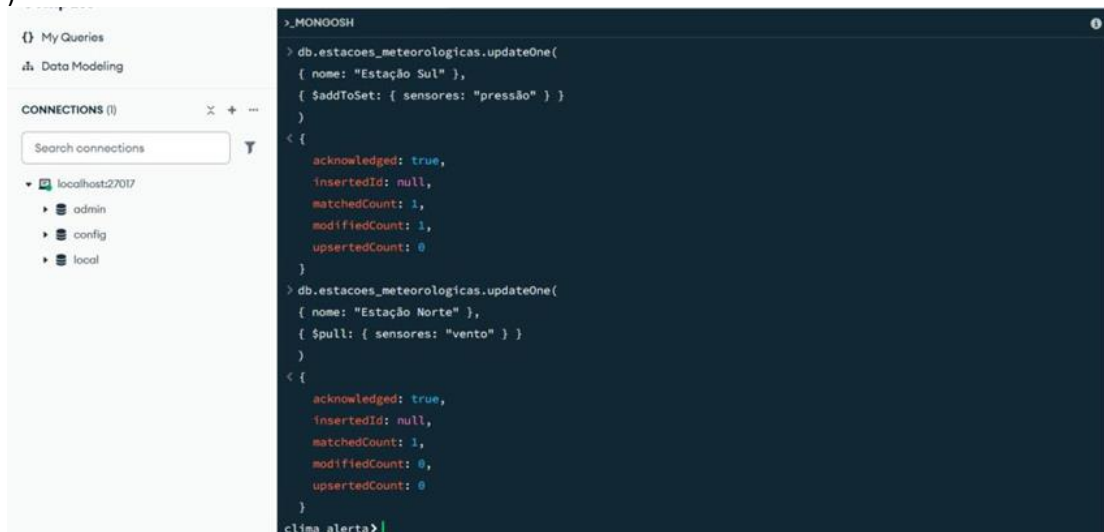
10)

```
db.estacoes_meteorologicas.updateOne(
```

```
{ nome: "Estação Norte" },
```

```
{ $pull: { sensores: "vento" } }
```

```
)
```



The screenshot shows the MongoDB Compass interface. The main editor displays two JavaScript snippets for update operations. The output shows the results of both operations, indicating that two documents were updated successfully.

```
> db.estacoes_meteorologicas.updateOne(
  { nome: "Estação Sul" },
  { $addToSet: { sensores: "pressão" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
> db.estacoes_meteorologicas.updateOne(
  { nome: "Estação Norte" },
  { $pull: { sensores: "vento" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
```

11)

```
db.estacoes_meteorologicas.deleteMany({
```

```

$or: [

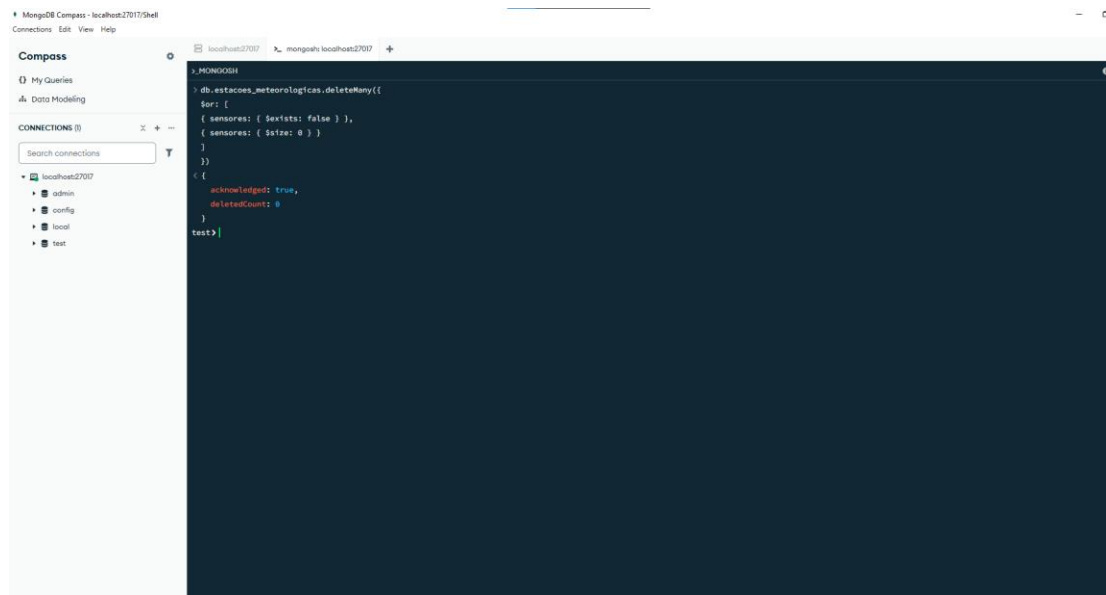
{ sensores: { $exists: false } },

{ sensores: { $size: 0 } }

]

})

```

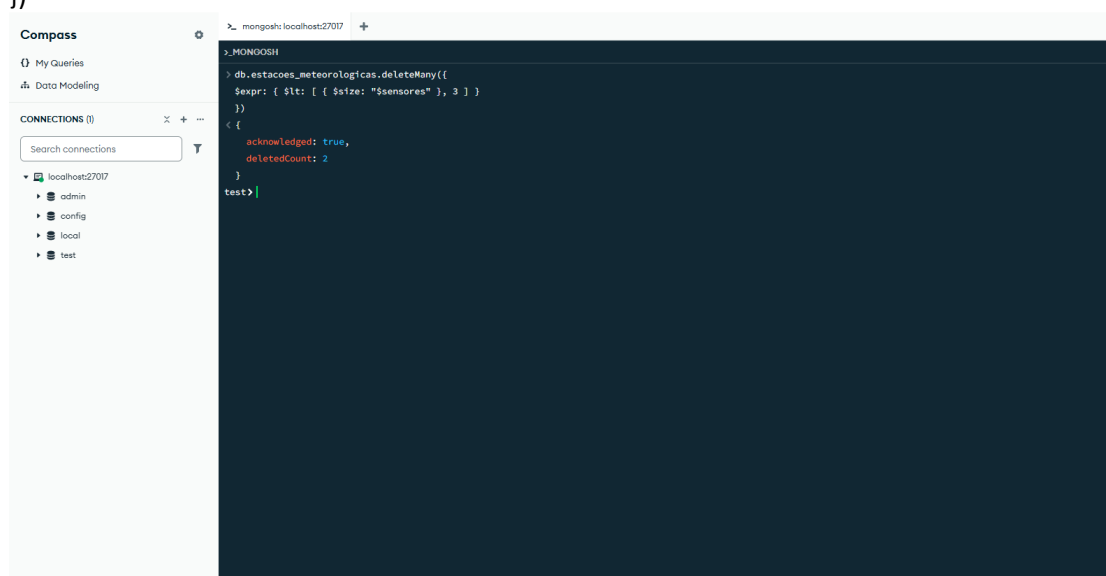


12)

```

db.estacoes_meteorologicas.deleteMany({
  $expr: { $lt: [ { $size: "$sensores" }, 3 ] }
})

```

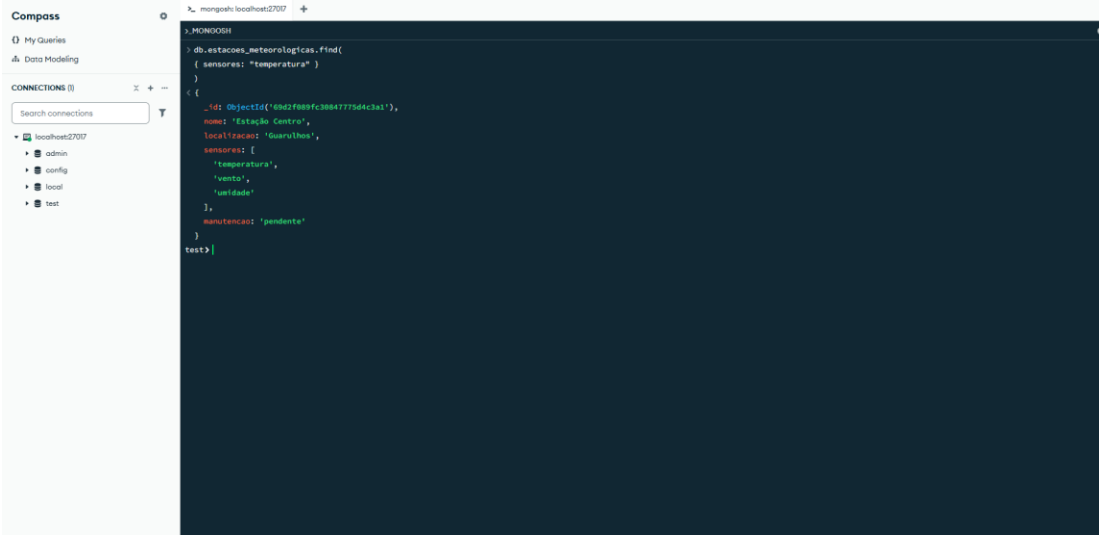


13)

```
db.estacoes_meteorologicas.find(
```

```
{ sensores: "temperatura" }
```

)

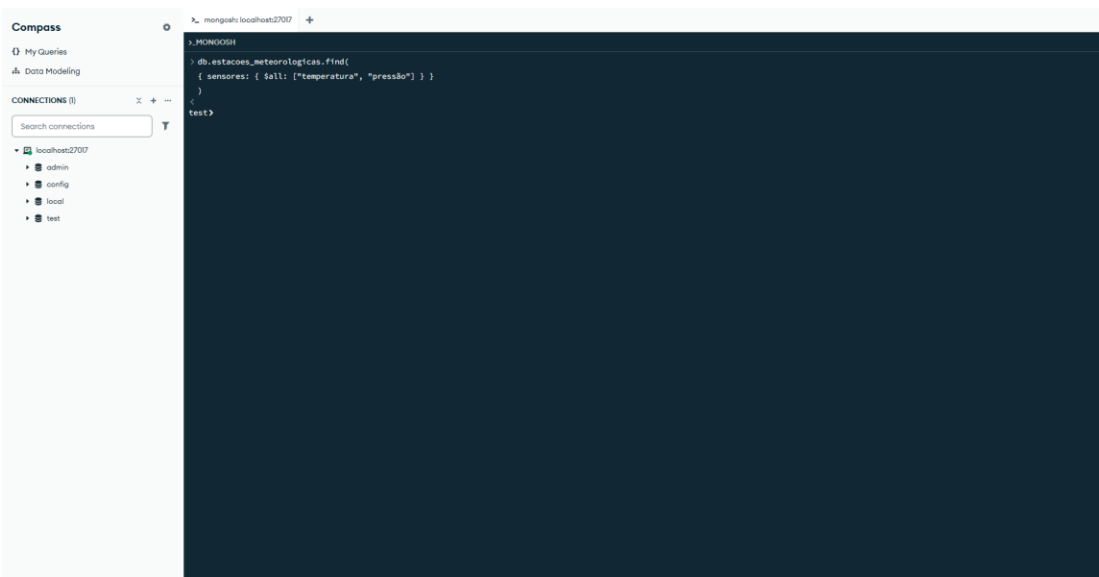


14)

```
db.estacoes_meteorologicas.find(
```

```
{ sensores: { $all: ["temperatura", "pressão"] } }
```

)

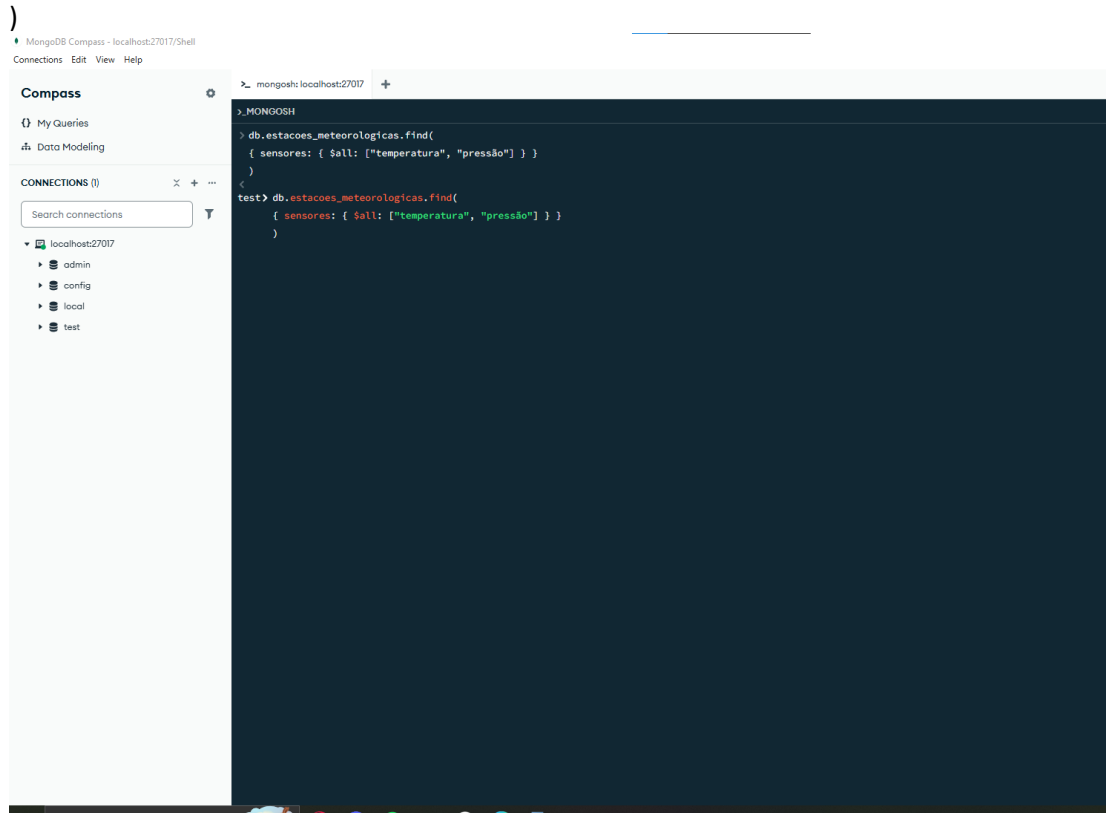


15)

```
db.estacoes_meteorologicas.find(
```

```
{ sensores: { $all: ["temperatura", "pressão"] } }
```





16)

`db.estacoes_meteorologicas.find().skip(2).limit(4)`

